AGENTIC ATSC LAB · STUDY GUIDE

What is still unsolved, and how to solve it

The research landscape from max-pressure to language-model traffic agents, the eleven gaps that keep them off real streets, five small SUMO experiments that make those gaps visible, and the designs and questions that look most promising.

Go 3 · Research

INSIDE

- Six waves of signal-control research, and why none of the newest has left the simulator
- How to tell a real improvement from noise, worked by hand
- Why a safety shield costs 1.6 s when the AI is perfect and saves 19 s when it is not
- Why promising drivers a light change can add delay and CO_2

Beginner-friendly companion to smrahmanbd.github.io/agentic-atsc-lab

The map: AATSC Go 3

This guide is for beginners who have met the earlier parts of the course (or are happy to learn the words as they go). You need only school maths. Go 3 is about research: what has been tried, how to judge it fairly, what is still missing, and what our own small experiments show. Every paper cited here is in the course's list of 40 checked references, and every lab number comes from this repository's SUMO runs. There are six ideas. Each builds on the one before.

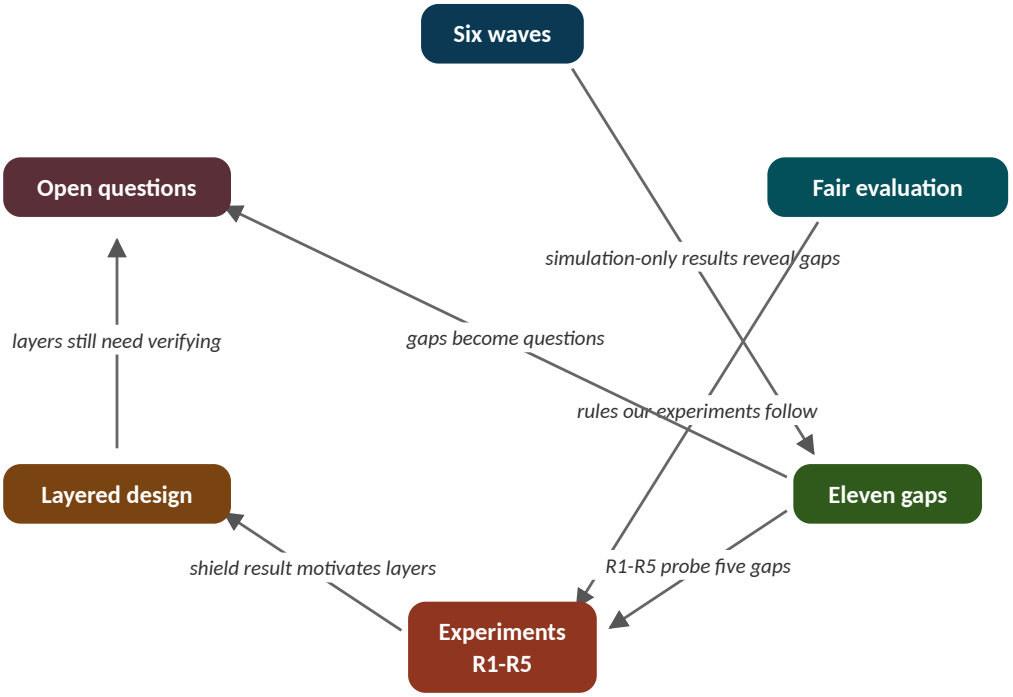


Figure 0. The whole guide on one page. Each box is one concept section; the arrows say how one idea feeds the next. Come back to this page whenever you feel lost.

1. The research landscape: six waves
2. How to evaluate a new idea fairly
3. The eleven gaps
4. Our five experiments, R1-R5
5. Future solution: a layered agentic controller
6. Open research questions

HOW TO USE THIS GUIDE

Read one concept per sitting (20 to 30 minutes). Do every **Simple calculation** with a pencil before reading the answer: most are percentages and averages. Say the **Picture this** scene out loud while you imagine walking through Research Park. The **Recap so far** box grows as you go; reading it is free revision. Finish with the self-quiz, then try to write one research question of your own using the last section.

What you will be able to do

- Name the six waves of signal-control research and say, in plain words, what reinforcement learning (RL) and a large language model (LLM) do.
- List the rules of a fair comparison and decide whether a difference is bigger than the noise.
- Explain at least six of the eleven gaps (sim-to-real, safety, hallucination, explainability, robustness, equity…) with one piece of evidence each.
- Read the results of experiments R1-R5 and compute relative changes from them.
- Sketch the layered agentic controller and say what each layer protects against.
- Turn a gap into a testable research question with a method, setting and metric.

Before we start

HOOK

Not one LLM traffic controller is on a real street. The nine language-model controllers in our reviewed literature were all tested in simulation only; none reports a field deployment. Their headline gains, about 17.5-20.4 % in waiting or travel time, are as reported by their authors.

HOOK

Breaking the rules looks faster. In our experiment R2, a perfectly reliable model with no safety shield had less delay (9.0 s) than the shielded one (10.6 s). It got there by issuing 112 rule-breaking commands. A fair comparison makes everyone obey the same rules.

HOOK

A promise can cost more than it saves. In R5, making the signal commit to its next change 8 s ahead gave connected cars usable speed advice and the fewest stops, but raised delay from 16.7 to 21.1 s and CO_2 from 229.5 to 241.4 g per vehicle.

Words you will meet

You do not need to memorise these now. Each one is explained again when it first matters.

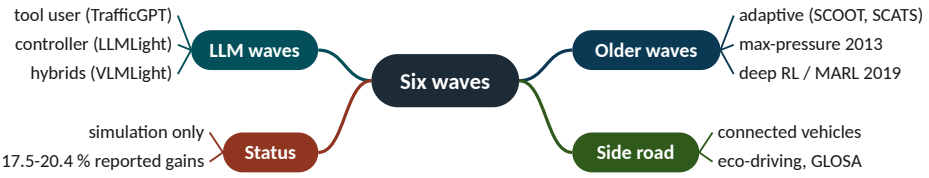
Word	In plain words
Reinforcement learning (RL)	A program that learns by trial and error, rewarded with a score, like a dog trained with treats.
Large language model (LLM)	A text-predicting AI, as in chatbots. It can answer 'switch to north-south' and give a reason.
Agent	A program that observes, decides and acts on its own. Here, a signal controller.
Hallucination	An LLM answer that is confident but wrong, such as a phase that does not exist.
Safety shield	A strict rule-checker that blocks any command breaking minimum green or other safety rules.
Sim-to-real gap	Everything that changes when a controller moves from a simulator (SUMO, CityFlow) to real streets.
Seed	The starting number for simulated random arrivals: a different seed is a different, equally likely day.
CAV / V2X	Connected and automated vehicle; V2X is the radio link between vehicles and the signal.
GLOSA	Green Light Optimal Speed Advisory: the speed that arrives on green, so the driver need not stop.

CONCEPT 1 OF 6 · AATSC GO 3

The research landscape: six waves

HOOK

Traffic lights have been 'smart' since the 1980s. So why are language-model controllers news, and why has none reached a real street?



IN PLAIN WORDS

Think of how phones grew up: landline, mobile, smartphone, voice assistant. Each new wave kept the old one inside it: a voice assistant still makes plain phone calls. Newer signal controllers likewise use older controllers as tools or as the yardstick. *Where the picture breaks:* every phone wave reached millions of real users; the newest signal-control waves have so far been tested only in simulators.

Adaptive systems (1980s on, SCOOT and SCATS) measure traffic and re-tune cycle, split and offset. **Max-pressure** (2013) gives green to the movement with the biggest 'queue in minus queue out'; it needs only local queues and has a mathematical proof of maximum throughput (Varaiya, 2013). Hardware-in-the-loop tests on real signal hardware move it toward the field (Stern et al., 2024).

Deep RL and multi-agent RL (MARL), from about 2019: a neural network learns which phase to pick by trial and error in a simulator, rewarded for less queueing. PressLight, CoLight, FRAP, AttendLight and MPLight are the standard set; MPLight ran on a simulated network of 2,510 lights.

LLM waves, from 2023. As a **tool user**, an LLM chats with an operator and calls simulators (TrafficGPT, Open-TI). As a **controller**, it reads the traffic state and picks the phase with a reason (LLMLight, LA-Light, CoLLMLight). **Hybrids** let fast RL or rules handle routine cases and send hard ones to LLM reasoning or critique (iLLM-TSC, VLMLight, HeraldLight). Reported gains: about 17.5-20.4 % against RL, in the authors' simulations; none of the nine reports a field deployment.

$$\text{new value} = \text{old value} \times (1 - \text{gain} \div 100)$$

how a reported percentage reduction works

$$\text{relative change} = (\text{new} - \text{old}) \div \text{old} \times 100\ \%$$

negative means a reduction

SIMPLE CALCULATION · What does '20.4 % less waiting' mean?

LA-Light reports 20.4 % less waiting than RL under sensor outages. Suppose (invented) RL waited 50 s.

1. Turn the percentage into a fraction: 20.4 ÷ 100 = **0.204**.
2. Seconds saved: 50 × 0.204 = **10.2 s**.
3. New waiting time: 50 − 10.2 = **39.8 s** (or 50 × 0.796).
4. Check backwards: (39.8 − 50) ÷ 50 × 100 = **−20.4 %**.
5. iLLM-TSC reports 17.5 % under degraded communication: 50 × 0.825 = **41.25 s**.

The two gains differ by 1.45 s here, and their setups differ: do not rank them.

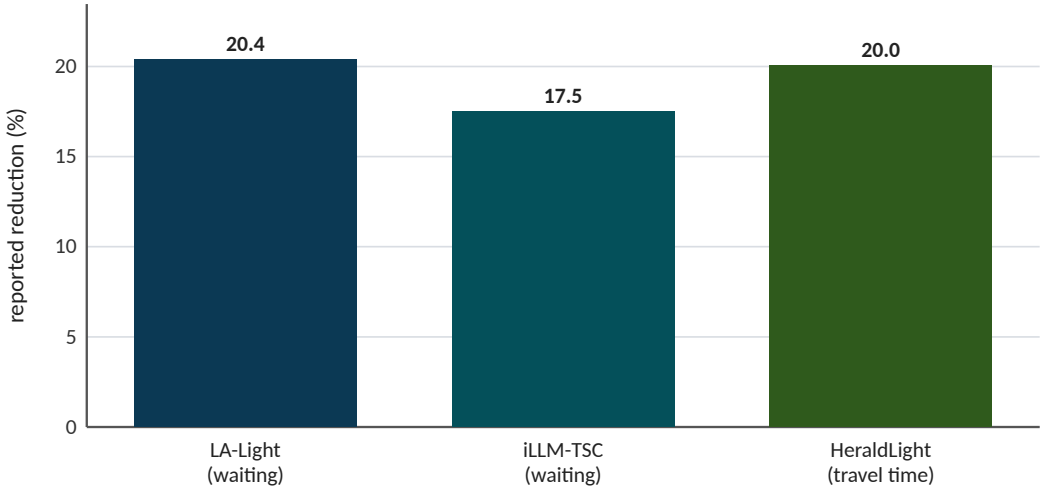


Figure 1. Headline gains reported by three LLM papers (authors' simulations). What to notice: all about one fifth, all in simulation, different setups.

TRAP

A reported percentage holds only for the authors' simulator, data and baseline. Two papers' percentages compare only if the setups match, which is rare.

PICTURE THIS · STOP 1 OF THE ROUTE

Six bone-dry surfboards: an old wooden SCOOT board, a pressure-gauge board, a neural-network board, a talking board, a board with a brain, a board split in two.
What it encodes: Adaptive → max-pressure → deep RL → LLM tool users → LLM controllers → hybrids; dry = simulation only.

TRY IT NOW (about 90 seconds)

VLMLight reports up to 65 % less emergency-vehicle waiting than RL alone. If (invented) an ambulance waited 40 s, how long after a 65 % cut? *Answer in the key at the back.*

RECAP SO FAR

1. **Six waves: adaptive → max-pressure → deep RL → LLM tool user → LLM controller → hybrids; LLM gains of about 17.5-20.4 % are reported in simulation only.**

CONCEPT 2 OF 6 · AATSC GO 3

How to evaluate a new idea fairly

HOOK

Your new controller cuts delay by 0.66 s. Is that a discovery, or just a lucky day of random traffic?



IN PLAIN WORDS

A fair race gives every runner the same shoes and track, runs more than one heat, and also checks who was left behind. One heat on a windy day proves little: the winner may just have had the wind at their back. *Where the picture breaks:* in a simulator we can replay exactly the same 'wind' (seed) for every controller, which real races cannot do.

Strong baselines. Always compare with well-tuned actuated control and max-pressure, not only fixed time, which is easy to beat. **Same everything else:** same network, demand, seeds, yellow and all-red, minimum green. In our code every controller drives the same Junction class, so results differ only by decisions.

Several seeds. A seed is the random starting number for arrivals; each seed is a different but equally likely day. Report the **mean** and the **spread** over at least 3 seeds. If a gain is smaller than the spread, you have not shown it. In R1 talking agents beat silent ones by 0.66 s, but the runs with messages wobble over a 0.67 s range: the effect is within single-seed noise.

More than delay. Report stops, CO_2, the 95th-percentile and the longest wait (fairness), and safety violations. **Stress tests** (faults, latency, odd demand, bad model outputs) and **ablations** (switch one part off at a time) show what really helps. And publish **negative results**: several of ours are. Rules matter too: in R2 the unshielded perfect model looked faster (9.02 s vs 10.62 s) only because it broke minimum green 112 times.

$$\text{mean} = \text{sum of results} \div \text{number of runs}$$

the typical value

$$\text{range} = \text{largest} - \text{smallest}$$

the simplest spread; the standard deviation is a finer one

SIMPLE CALCULATION · R1: is the gain bigger than the wobble?

R1, four-junction corridor, one seed. No messages: 18.45 s. Six runs with messages (latency 0, 3, 6, 10 s; loss 30 %, 60 %): 17.79, 17.69, 17.63, 17.91, 18.30, 17.86 s. Latency and loss can only hurt, so up-and-down changes behave like noise.

1. Sum: 17.79 + 17.69 + 17.63 + 17.91 + 18.30 + 17.86 = **107.18 s**.
2. Mean: 107.18 ÷ 6 = **17.86 s**.
3. Range: 18.30 − 17.63 = **0.67 s** (standard deviation about 0.24 s).
4. Benefit of talking at 0 s latency: 18.45 − 17.79 = **0.66 s**, that is 3.6 %.
5. Compare: the benefit (0.66 s) is no bigger than the wobble (0.67 s).

With one seed we cannot claim that messages help, or that latency hurts. The honest next step is more seeds, longer corridors and heavier demand.

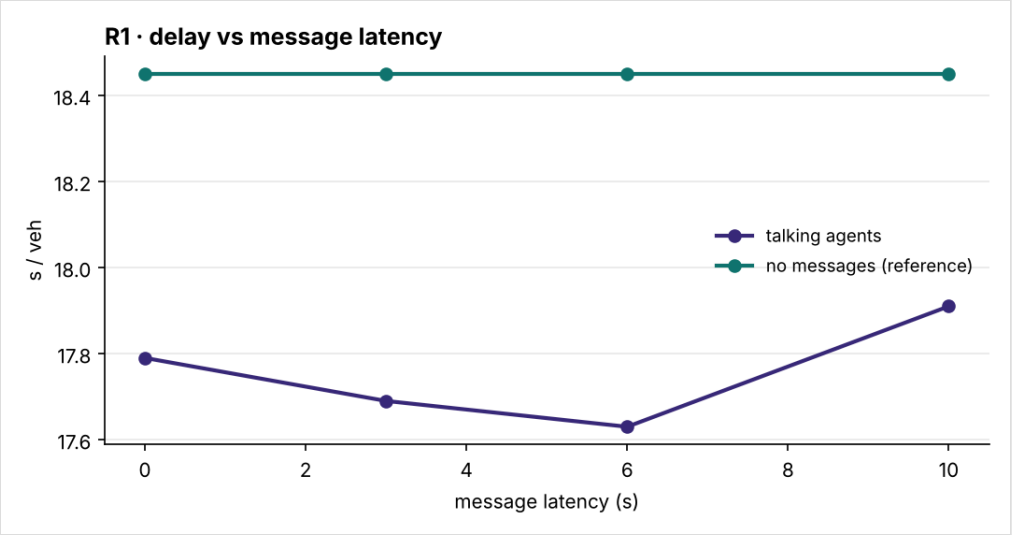


Figure 2. R1: corridor delay against message latency (one seed). What to notice: the vertical axis spans only 17.6-18.4 s. Delay goes down then up as latency grows, which no theory predicts: it is noise. All talking runs sit under the no-message line by less than 1 s.

TRAP

Do not zoom into a chart and read every wiggle as an effect. Check the axis: R1's whole chart covers less than 1 s. A difference smaller than the run-to-run spread is not a result, however neat the line looks.

PICTURE THIS · STOP 2 OF THE ROUTE

Referees make every runner use the same shoes, run three races, and time not just speed but also who was left behind.

What it encodes: Same shoes = same rules; three races = seeds; left behind = fairness metrics.

TRY IT NOW (about 90 seconds)

Practice numbers (invented): a new controller gives 10, 11 and 12 s delay over three seeds; the old one gives 11.5, 12 and 12.5 s. Find both means and ranges. Is the gain convincing? *Answer in the key at the back.*

RECAP SO FAR

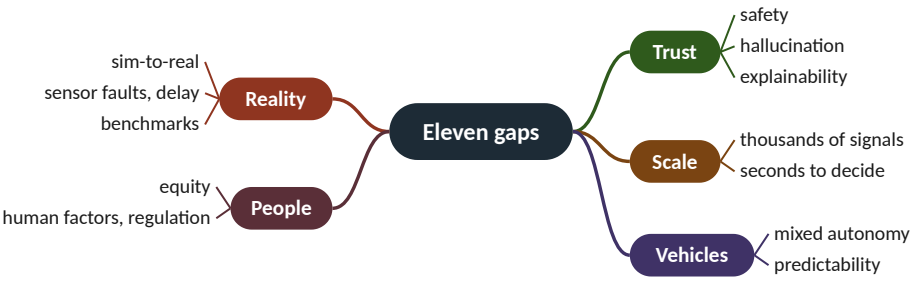
1. **Six waves:** adaptive → max-pressure → deep RL → LLM tool user → LLM controller → hybrids; LLM gains of about 17.5-20.4 % are reported in simulation only.
2. **Fair evaluation: strong baselines, same rules, 3+ seeds with mean and spread, metrics beyond delay, stress tests, ablations, honest negatives.**

CONCEPT 3 OF 6 · AATSC GO 3

The eleven gaps

HOOK

If LLM and RL controllers report 20 % gains, why is no city using them? Because eleven potholes lie between a good simulation result and a safe street.



IN PLAIN WORDS

A new car that wins on a test track still needs crash tests, rain tests, cheap fuel, a driver who understands the dashboard and a road-safety licence. Speed on the track is only the first step. *Where the picture breaks:* car makers have an agreed licensing process; for AI signal controllers no such certification route exists yet.

Sim-to-real: almost every controller is tested only in simulators; Sim2Signal finds direct transfer always degrades and no fix wins everywhere. **Safety:** minimum greens and conflict rules are non-negotiable, but learning and language models do not respect them by default. **Hallucination:** an LLM can name an invalid phase or give a confident, wrong reason, and its response time is unbounded. **Explainability:** operators must understand why a phase was chosen.

Sensor faults and communication delay: detectors fail and radio links drop messages, yet most benchmarks assume perfect data. **Scalability and latency:** cities have thousands of signals deciding every few seconds. **Fairness and equity:** throughput-optimal control can starve side streets, pedestrians and bus riders. **Data and benchmarks:** results rest on a few data sets (Jinan, Hangzhou, New York).

Mixed autonomy: connected-vehicle benefits depend on how many vehicles are connected; Feng et al. (2015) report up to 16.33 % less delay than actuated control at high penetration but about the same at low. **Adaptivity vs predictability:** eco-driving needs to know when the light changes. **Human factors and regulation:** trust, advice compliance and certification decide deployment, and are under-researched. Our R4 measures the equity gap on one junction.

$$\text{fairness-weighted score} = \text{efficiency term} + w \times \text{longest wait}$$

R4 sweeps the weight w from 0 to 1

$$\text{relative change} = (\text{new} - \text{old}) \div \text{old} \times 100\ \%$$

used for each metric separately

SIMPLE CALCULATION · R4: what does fairness cost?

One junction, heavy main road, one seed. Weight w = 0: mean delay 9.69 s, longest wait 53 s. w = 0.15: 9.56 s, 22 s. w = 0.5: 10.46 s, 16 s.

1. Longest wait, w 0 → 0.15: (22 − 53) ÷ 53 × 100 = **−58.5 %**.
2. Mean delay, w 0 → 0.15: (9.56 − 9.69) ÷ 9.69 × 100 = **−1.3 %** (no cost at all).
3. Longest wait, w 0.15 → 0.5: (16 − 22) ÷ 22 × 100 = **−27.3 %**.
4. Mean delay, w 0.15 → 0.5: (10.46 − 9.56) ÷ 9.56 × 100 = **+9.4 %**.
5. Max-pressure (ignores waits): longest wait 150 s, about 150 ÷ 22 = **6.8 times** the w = 0.15 agent.

A small fairness weight was free in this run; larger weights buy a little more fairness for extra delay. One seed only, so treat the 'free' part as a hint, not a proof.

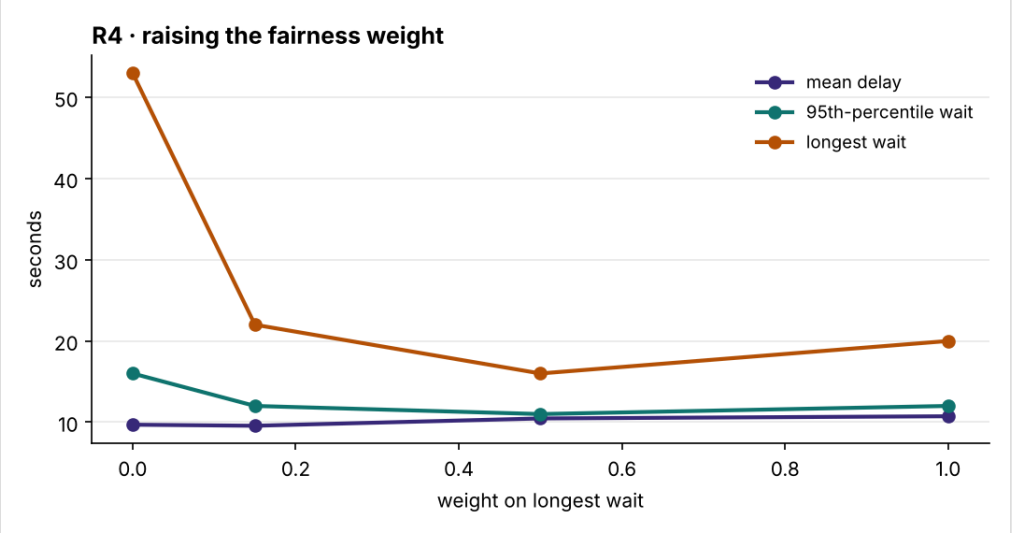


Figure 3. R4: raising the fairness weight (one junction, one seed). What to notice: the longest wait (top line) falls from 53 s to 22 s at w = 0.15 while mean delay (bottom line) barely moves. After that, mean delay creeps up and the longest wait no longer improves much (20 s at w = 1).

TRAP

A low mean delay can hide one driver waiting a very long time. Max-pressure had more mean delay here (15.52 s), but the bigger warning is its 150 s longest wait. Always report the tail (95th percentile and maximum), not just the average.

PICTURE THIS · STOP 3 OF THE ROUTE

Eleven potholes, each with a sign: a simulator screen, a broken guardrail, a parrot, a question mark, a cut cable, a crowd, a seesaw, blue cars, a broken promise, a single dataset, a judge's gavel.
What it encodes: Sim-to-real, safety, hallucination, explainability, robustness, scale and latency, equity, mixed autonomy, predictability, benchmarks, human factors and regulation.

TRY IT NOW (about 90 seconds)

Jayawardana and Wu report up to 25 % less CO_2 from RL eco-driving at full CAV penetration, and that 25 % penetration delivers at least half of the benefit. What CO_2 cut does that imply at 25 % penetration, at least? *Answer in the key at the back.*

RECAP SO FAR

1. **Six waves:** adaptive → max-pressure → deep RL → LLM tool user → LLM controller → hybrids; LLM gains of about 17.5-20.4 % are reported in simulation only.
2. **Fair evaluation:** strong baselines, same rules, 3+ seeds with mean and spread, metrics beyond delay, stress tests, ablations, honest negatives.
3. **Eleven gaps: sim-to-real, safety, hallucination, explainability, robustness, scale and latency, equity, mixed autonomy, predictability, benchmarks, human factors. R4: w = 0.15 cut the longest wait 53 → 22 s for no delay cost.**

CONCEPT 4 OF 6 · AATSC GO 3

Our five experiments, R1–R5

HOOK

What happens when the AI that runs the lights gives a bad answer four times in ten? We can test it safely in SUMO, and the answer depends entirely on one guard.



IN PLAIN WORDS

A seat belt slightly annoys you on every trip and does nothing on most of them. On the day of a crash it is the only thing that matters. The safety shield is the seat belt between an LLM and the lights. *Where the picture breaks:* a seat belt never makes the trip slower; the shield sometimes overrules a good-but-early switch, which costs a little delay.

R1, latency and loss (four-junction corridor): talking agents with messages delayed 0-10 s or 30-60 % lost. Delay 17.6-18.3 s against 18.45 s without messages: under 1 s benefit, and latency and loss effects within single-seed noise. **R2, hallucination and shield** (one junction): a **mock** LLM gives a bad answer with probability h. With the shield, bad answers are caught and replaced.

In R2 at h = 40 %, the shielded agent keeps delay at **12.14 s** (longest wait 39 s, 0 violations); without the shield, delay is **31.49 s**, the longest wait 132 s, and **223** unsafe commands reach the signal. **R3, CAV-only sensing**: the agent sees only connected vehicles and scales up by the penetration. At 10 % delay is 9.94 s vs 9.01 s with loops; at 100 % both are 6.6 s.

R4, fairness: a small weight on the longest wait (0.15) cut it from 53 to 22 s and slightly lowered mean delay. **R5, predictability** (corridor, 50 % CAVs with GLOSA): committing H seconds ahead gives the fewest stops at H = 8 s (0.86) but raises delay and CO_2. Every run is one seed in a small synthetic scenario: illustrations, not proofs.

$$\text{relative change} = (\text{with} - \text{without}) \div \text{without} \times 100\ \%$$

for the shield, 'without' is the unshielded run

$$\text{times worse} = \text{unshielded} \div \text{shielded}$$

a ratio, another way to say the same thing

SIMPLE CALCULATION · R2: the shield at 40 % bad answers

One junction, mock LLM, 40 % bad answers, one seed. With shield: delay 12.14 s, longest wait 39 s. Without: delay 31.49 s, longest wait 132 s.

1. Delay saved: 31.49 − 12.14 = **19.35 s** per vehicle.
2. As a share of the unshielded delay: 19.35 ÷ 31.49 × 100 = **61.4 %** less.
3. Ratio: 31.49 ÷ 12.14 = **2.59**: without the shield, delay is about 2.6 times higher.
4. Longest wait: 132 − 39 = **93 s** shorter with the shield.
5. Shielded agent, 0 % → 40 % bad: (12.14 − 10.62) ÷ 10.62 × 100 = **+14.3 %** only.

With the shield, a model that is wrong 40 % of the time costs only 14 % more delay and breaks no rules. Without it, the junction degrades badly and 223 unsafe commands get through.

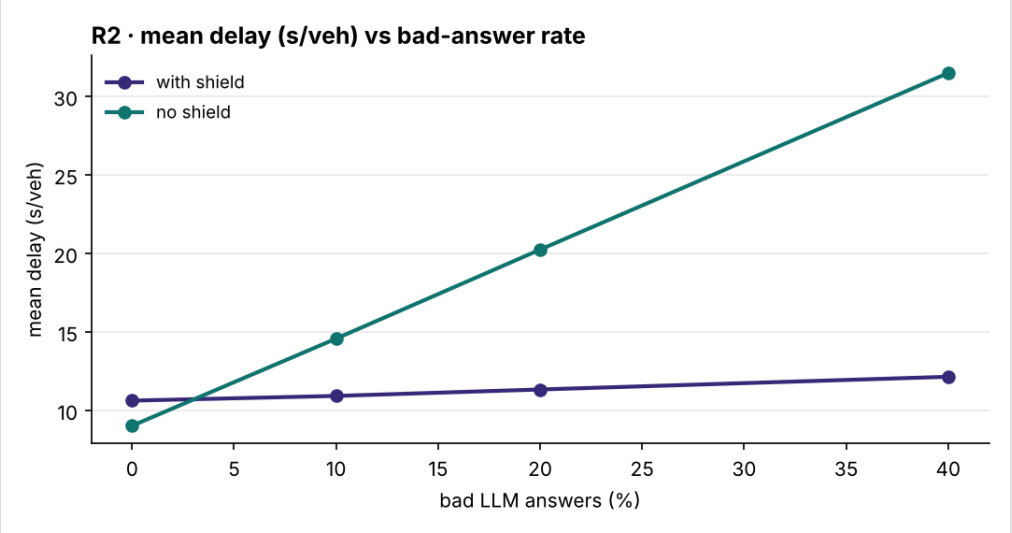


Figure 4. R2: mean delay against the share of bad LLM answers (mock LLM, one seed). What to notice: the shielded line is almost flat (10.6 → 12.1 s); the unshielded line climbs steeply (9.0 → 31.5 s). At 0 % the unshielded agent looks faster, but only because it breaks minimum green.

TRAP

R2 tests the **shield**, not any real language model: the 'LLM' is a mock that is wrong with a set probability. Do not quote these numbers as how ChatGPT-like models behave at junctions.

PICTURE THIS · STOP 4 OF THE ROUTE

Room 1: tin-can phones with cut strings. Room 2: the parrot without its guard wrecks the junction. Room 3: half-blind goggles. Room 4: a seesaw between speed and fairness. Room 5: a clock that keeps promises.

What it encodes: R1 latency/loss, R2 shield vs hallucination, R3 CAV-only sensing, R4 fairness weight, R5 commitment horizon for eco-driving.

TRY IT NOW (about 90 seconds)

In R3 at 10 % connected vehicles, the CAV-only agent had 9.94 s delay and the loop-detector agent 9.01 s. What is the relative change? At 100 % both had 6.6 s: what is it then? *Answer in the key at the back.*

RECAP SO FAR

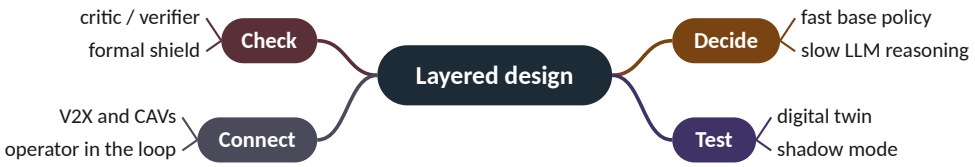
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3. **Eleven gaps:** sim-to-real, safety, hallucination, explainability, robustness, scale and latency, equity, mixed autonomy, predictability, benchmarks, human factors. R4: w = 0.15 cut the longest wait 53 → 22 s for no delay cost.
4. **Experiments R1-R5: messages help under 1 s (noise); the shield holds delay at 12.1 s vs 31.5 s with 40 % bad answers; CAV-only works but degrades at low penetration; a little fairness is free; commitment trades stops for delay.**

CONCEPT 5 OF 6 · AATSC GO 3

Future solution: a layered agentic controller

HOOK

Should one giant AI model run every traffic light in a city? The evidence says no: the most promising designs look like a layer cake.



IN PLAIN WORDS

A hospital does not send every patient to the top surgeon. Nurses handle routine cases fast; hard cases go to a specialist; a second doctor checks risky decisions; strict rules forbid dangerous doses; the hospital board sets the goals. *Where the picture breaks:* a surgeon can take an hour; a signal must decide within seconds, so the slow layer must be optional, never on the critical path.

Fast base policy: max-pressure, actuated logic or a small RL policy decides every few seconds. Cheap, stable and already proven. **Slow reasoning agent:** an LLM is consulted only for unusual cases (incidents, emergency vehicles, special events) or to audit the base policy. VLMLight does this and reports under 1 % degradation in normal conditions; iLLM-TSC lets an LLM correct RL actions.

Critic and verifier: a second model or a rule checker reviews every proposal for invalid or hallucinated actions (HeraldLight's LLM-Critic). **Formal safety shield:** hard constraints in the signal cabinet (minimum green, intergreen, pedestrian times, conflict monitor), never bypassed. Our R2 shows why: it held delay at 12.14 s with 40 % bad answers.

Digital twin: a calibrated simulation of the corridor where new policies run in **shadow mode** (deciding, but not controlling) before going live, and where sim-to-real gaps are measured. **V2X and CAVs:** signal timing broadcasts with commitment windows for eco-driving, and CAV trajectories as a sensor. **Operator in the loop:** engineers state goals in plain language, approve autonomy levels and read the agents' explanations.

$$\text{decisions per hour} = \text{signals} \times (3600 \div \text{seconds between decisions})$$

the load on the controller

$$\text{LLM calls per hour} = \text{decisions per hour} \times \text{share sent to the slow layer}$$

why escalation must be rare

SIMPLE CALCULATION · Why not ask the LLM every time?

MPLight was demonstrated on 2,510 simulated lights. Suppose (invented settings) each decides every 5 s, and only 1 % of decisions are escalated to the LLM.

1. Decisions per light per hour: 3600 ÷ 5 = **720**.
2. Whole network: 2,510 × 720 = **1,807,200** decisions per hour.
3. Per second: 1,807,200 ÷ 3600 = **502** decisions every second.
4. Escalate only 1 %: 1,807,200 × 0.01 = **18,072** LLM calls per hour.
5. That is about 18,072 ÷ 3600 = **5** LLM calls per second instead of 502.

The fast layer handles 99 % of the work; the slow layer gets 100 times fewer calls. A learned 'when to escalate' rule is an open research question.

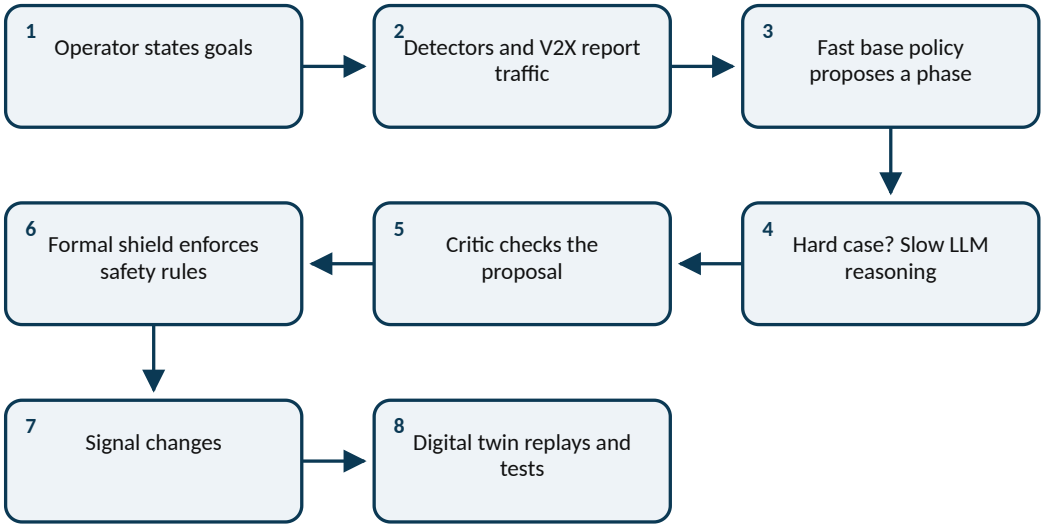


Figure 5. The layered agentic controller. What to notice: the fast base policy makes routine decisions; the LLM is consulted only for hard cases; every proposal passes the critic and then the shield before reaching the lights. The digital twin tests changes first, and people set the goals.

TRAP

The shield is not the critic. The critic is a model or checker that can be wrong and gives advice; the shield is fixed, hard-coded rules in the signal cabinet that can never be bypassed. A safe design needs the shield even if the critic is excellent.

PICTURE THIS · STOP 5 OF THE ROUTE

A seven-layer cake: a fast sponge base, a slow thinking cream, a taster who checks every slice, an iron plate underneath, a mirror cake (digital twin) beside it, radio candles on top, and the chef watching. *What it encodes:* Fast base policy, slow LLM reasoning, critic, formal shield, digital twin, V2X, operator in the loop.

TRY IT NOW (about 90 seconds)

A city has 400 signals deciding every 4 s. How many decisions per hour? If 2 % are escalated to an LLM, how many LLM calls per hour? *Answer in the key at the back.*

RECAP SO FAR

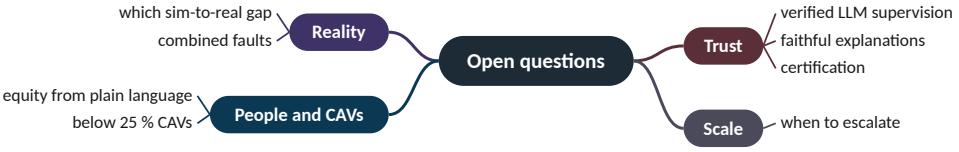
1. **Six waves:** adaptive → max-pressure → deep RL → LLM tool user → LLM controller → hybrids; LLM gains of about 17.5-20.4 % are reported in simulation only.
2. **Fair evaluation:** strong baselines, same rules, 3+ seeds with mean and spread, metrics beyond delay, stress tests, ablations, honest negatives.
3. **Eleven gaps:** sim-to-real, safety, hallucination, explainability, robustness, scale and latency, equity, mixed autonomy, predictability, benchmarks, human factors. R4: w = 0.15 cut the longest wait 53 → 22 s for no delay cost.
4. **Experiments R1-R5:** messages help under 1 s (noise); the shield holds delay at 12.1 s vs 31.5 s with 40 % bad answers; CAV-only works but degrades at low penetration; a little fairness is free; commitment trades stops for delay.
5. **Layered design: fast base policy, slow LLM for rare hard cases, critic, formal shield, digital twin (shadow mode), V2X, operator in the loop.**

CONCEPT 6 OF 6 · AATSC GO 3

Open research questions

HOOK

R5 gave drivers a promise about the next light change and got fewer stops, but more delay and more CO_2. Nobody yet knows how to get predictability without losing adaptivity. That is what an open question looks like.



IN PLAIN WORDS

A lighthouse does not show you the island; it shows where the rocks are so you can sail toward it. A good research question works the same way: it marks clearly what is not known, so a study can head straight for it. *Where the picture breaks:* a lighthouse beam is fixed; research questions change as soon as someone answers part of them.

Eight open questions. (1) **Verified LLM supervision:** can an LLM supervising max-pressure or RL be guaranteed never to break minimum green or a maximum pedestrian wait, and at what cost? (2) **Faithful explanations:** do an LLM's stated reasons match what really drove the decision? (3) **Which sim-to-real gap dominates** (observation, action, transition or reward) for LLM vs RL controllers? (4) **Combined faults:** detector outages, V2X latency and noisy prompts in one stress test.

(5) **Scalable escalation:** when should a cheap local policy call expensive reasoning under a fixed compute budget? (6) **Equity from plain language:** can operators state people-centric goals in words and compile them safely into constraints? (7) **Low-penetration CAVs:** below about 25 % connected vehicles, can joint signal and eco-driving control beat actuated control? (8) **Certification:** what evidence would a road authority need?

Our R5 shows why (7) and predictability are hard. With 50 % CAVs and GLOSA on the corridor, committing H seconds ahead gives the fewest stops at H = 8 s (0.86 vs 0.94), because more connected cars get usable advice, but delay rises from 16.72 to 21.12 s and CO_2 from 229.5 to 241.4 g. In our implementation the lost adaptivity costs more than eco-driving saves. A recipe for a question: **gap + method + setting + metric**, compared with actuated control and max-pressure, over 3+ seeds.

$$\text{research question} = \text{gap} + \text{method} + \text{setting} + \text{metric}$$

compared with actuated and max-pressure

$$\text{relative change} = (\text{H value} - \text{H 0 value}) \div \text{H 0 value} \times 100\ \%$$

R5 compares each horizon with fully adaptive control

SIMPLE CALCULATION · R5: the price of a promise

Corridor, 50 % CAVs with GLOSA, one seed. H = 0 (fully adaptive): delay 16.72 s, CO_2 229.5 g, 0.942 stops. H = 8 s: 21.12 s, 241.4 g, 0.860 stops. H = 16 s: 30.10 s, 262.3 g.

1. Stops, H 0 → 8: (0.860 − 0.942) ÷ 0.942 × 100 = **−8.7 %** (the gain).
2. Delay, H 0 → 8: 21.12 − 16.72 = 4.40 s; 4.40 ÷ 16.72 × 100 = **+26.3 %**.
3. CO_2, H 0 → 8: 241.4 − 229.5 = 11.9 g; 11.9 ÷ 229.5 × 100 = **+5.2 %**.
4. CO_2, H 0 → 16: (262.3 − 229.5) ÷ 229.5 × 100 = **+14.3 %**.
5. Delay, H 0 → 16: (30.10 − 16.72) ÷ 16.72 × 100 = **+80.0 %**.

An 8.7 % cut in stops cost 26 % more delay and 5 % more CO_2. Longer promises are worse on every count. Predictability without losing adaptivity remains open.

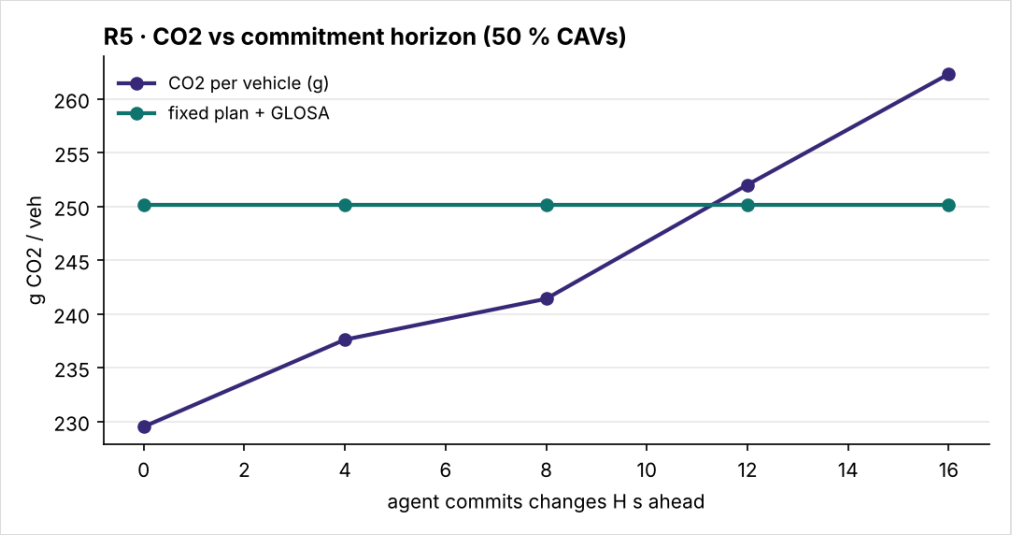


Figure 6. R5: CO2 per vehicle against the commitment horizon H (50 % CAVs, one seed). What to notice: CO2 rises steadily as the agent commits further ahead (229.5 g at H = 0 to 262.3 g at H = 16 s), and passes the fixed plan with GLOSA (250.1 g) at about H = 12 s.

TRAP

Fewer stops does not mean less CO_2 or less delay. In R5 the 8 s promise gave the fewest agent stops, yet both delay and CO_2 went up. Judge eco-driving by every metric, not the one it was designed to improve.

PICTURE THIS · STOP 6 OF THE ROUTE

A lighthouse sends eight beams into the fog, each lighting a different unexplored island. On the nearest, a promise-keeping clock stands by a car puffing grey smoke.
What it encodes: Eight beams = eight open questions; clock and smoke = R5's commitment raising CO_2.

TRY IT NOW (about 90 seconds)

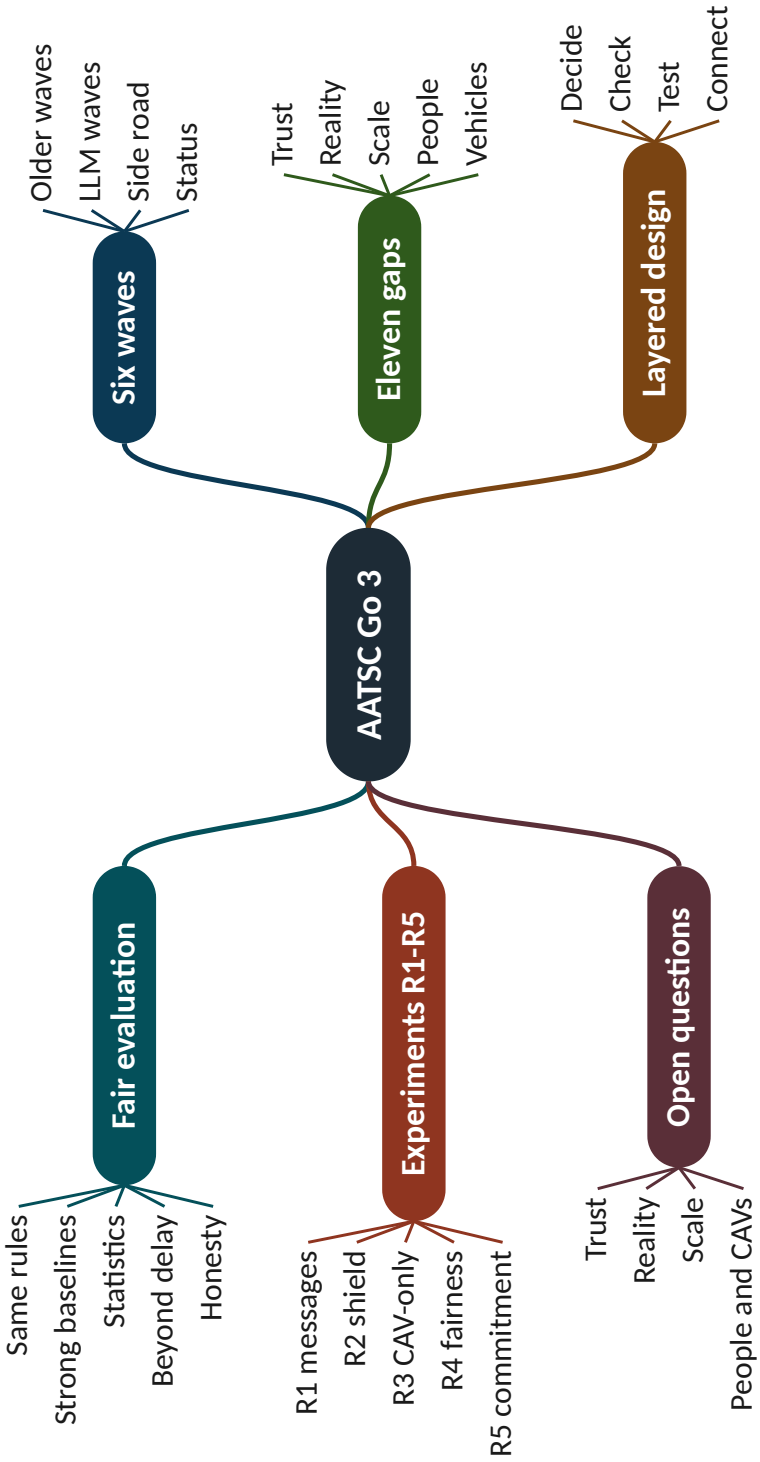
Build one research question for the 'low CAV penetration' gap: choose a method, a setting and a metric. Then compute R5's H = 4 s delay change: delay 18.87 s vs 16.72 s at H = 0. *Answer in the key at the back.*

RECAP SO FAR

1. **Six waves:** adaptive → max-pressure → deep RL → LLM tool user → LLM controller → hybrids; LLM gains of about 17.5-20.4 % are reported in simulation only.
2. **Fair evaluation:** strong baselines, same rules, 3+ seeds with mean and spread, metrics beyond delay, stress tests, ablations, honest negatives.
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4. **Experiments R1-R5:** messages help under 1 s (noise); the shield holds delay at 12.1 s vs 31.5 s with 40 % bad answers; CAV-only works but degrades at low penetration; a little fairness is free; commitment trades stops for delay.
5. **Layered design:** fast base policy, slow LLM for rare hard cases, critic, formal shield, digital twin (shadow mode), V2X, operator in the loop.
6. **Open questions: verified supervision, faithful explanations, dominant sim-to-real gap, combined faults, scalable escalation, equity, low-penetration CAVs, certification. R5: commitment cut stops 8.7 % but added 26 % delay.**

The master mind map

Every concept and its main branches on one page. Turn the page sideways. Try redrawing it from memory on blank paper, then compare.



Connections and contrasts

Four families of signal controller

	Max-pressure	Deep RL	LLM controller	Hybrid
Decides with	local queues in minus out	a policy learned from reward	a prompt describing traffic	fast policy + LLM for hard cases
Explains itself?	rule is simple	usually no	yes, a stated reason	yes, for escalated cases
Main worry	starving side streets	sim-to-real transfer	hallucination, latency	when to escalate
Strongest evidence	proof of max throughput; hardware-in-the-loop	simulation, up to 2,510 lights	simulation only	simulation only

Things beginners mix up

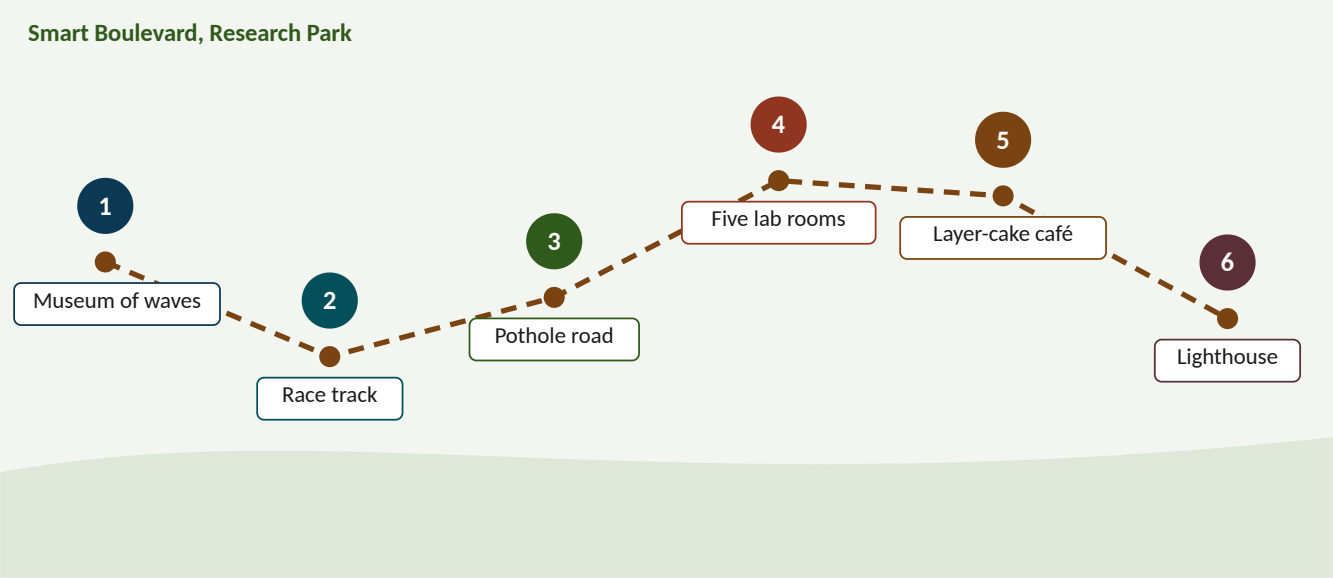
This…	…is not this	The one fact that tells them apart
RL	**LLM**	RL learns from reward by trial and error; an LLM predicts text and is prompted, not trained per junction.
Critic	**Shield**	A critic is a model giving advice and can be wrong; a shield is fixed rules that are never bypassed.
Seed	**Scenario**	A seed is another random day of the same traffic; a scenario is a different network or demand.
Mean delay	**Longest wait**	R4: mean delay barely moved while the longest wait fell 53 → 22 s.
Fewer stops	**Less CO2**	R5 at H = 8 s: fewest stops, but CO2 rose 229.5 → 241.4 g.
Reported gain	**Field result**	17.5-20.4 % LLM gains are authors' simulation results; none is from a real street.

Symbols and terms

Symbol / term	Meaning
RL / MARL	reinforcement learning; multi-agent RL with one learner per junction
LLM	large language model: predicts text, can give a reason
Hallucination	confident but wrong LLM output, e.g. an invalid phase
Shield	hard rule layer that blocks unsafe commands
Critic	second model or checker reviewing proposals
Sim-to-real gap	loss of performance moving from simulator to street: observation, action, transition, reward
Seed	random starting number for simulated arrivals
Ablation	switching off one component to see its effect
CAV, V2X	connected and automated vehicle; vehicle-to-everything radio
SPaT, GLOSA	signal phase and timing message; green light optimal speed advisory
Commitment horizon H	seconds ahead the signal promises its next change (R5)
Digital twin / shadow mode	calibrated simulation; running a policy without letting it control

The route: Smart Boulevard, Research Park

Walk on from Workshop Lane into Research Park, from a museum of surfboards to a lighthouse in the fog: one odd, loud scene per concept. Picture each one big, physical and silly.



The memory route. Walk it in order; each stop holds one concept.

Stop 1: Museum of waves (The research landscape: six waves). Six bone-dry surfboards: an old wooden SCOOT board, a pressure-gauge board, a neural-network board, a talking board, a board with a brain, a board split in two. *Adaptive → max-pressure → deep RL → LLM tool users → LLM controllers → hybrids; dry = simulation only.*

Stop 2: Race track (How to evaluate a new idea fairly). Referees make every runner use the same shoes, run three races, and time not just speed but also who was left behind. *Same shoes = same rules; three races = seeds; left behind = fairness metrics.*

Stop 3: Pothole road (The eleven gaps). Eleven potholes, each with a sign: a simulator screen, a broken guardrail, a parrot, a question mark, a cut cable, a crowd, a seesaw, blue cars, a broken promise, a single dataset, a judge's gavel. *Sim-to-real, safety, hallucination, explainability, robustness, scale and latency, equity, mixed autonomy, predictability, benchmarks, human factors and regulation.*

Stop 4: Five lab rooms (Our five experiments, R1-R5). Room 1: tin-can phones with cut strings. Room 2: the parrot without its guard wrecks the junction. Room 3: half-blind goggles. Room 4: a seesaw between speed and fairness. Room 5: a clock that keeps promises. *R1 latency/loss, R2 shield vs hallucination, R3 CAV-only sensing, R4 fairness weight, R5 commitment horizon for eco-driving.*

Stop 5: Layer-cake café (Future solution: a layered agentic controller). A seven-layer cake: a fast sponge base, a slow thinking cream, a taster who checks every slice, an iron plate underneath, a mirror cake (digital twin) beside it, radio candles on top, and the chef watching. *Fast base policy, slow LLM reasoning, critic, formal shield, digital twin, V2X, operator in the loop.*

Stop 6: Lighthouse (Open research questions). A lighthouse sends eight beams into the fog, each lighting a different unexplored island. On the nearest, a promise-keeping clock stands by a car puffing grey smoke. *Eight beams = eight open questions; clock and smoke = R5's commitment raising CO_2.*

WALK-THROUGH DRILL

Close the guide. Walk the route in your head, saying each stop out loud and the one number or formula it holds. Then check against the cheat sheet. Repeat until you can do it in under two minutes.

Self-quiz

18 questions: warm-up (recall), core (use the ideas), challenge (combine them). Answer on paper first, then check the key.

Warm-up

Q1. Signal-control research has moved in waves, each wave keeping the previous one as its baseline. Which of these waves came first?

A. Deep RL
B. Max-pressure
C. Adaptive systems such as SCOOT and SCATS
D. LLM controllers

Q2. Researchers list hallucination as a key barrier for language-model controllers. In plain words, what is an LLM 'hallucination' at a junction?

A. A slow answer
B. A confident but wrong answer, such as a phase that does not exist
C. A message lost on the radio link
D. A detector that stops working

Q3. A seed is the starting number for the simulator's random arrivals. Why should a study report its results over several seeds rather than one?

A. Seeds make the simulation faster
B. Each seed is a different random day of traffic, so you can see the spread
C. Seeds change the road network
D. Reviewers like more tables

Q4. Nine LLM traffic-signal controller papers were reviewed for this course. How many of them report a deployment on real streets, rather than only results from a simulator such as CityFlow or SUMO, at the time of the course's literature check?

A. None
B. One
C. Three
D. All nine

Q5. The layered agentic controller has seven layers, from the fast base policy to the operator. Which layer can never be bypassed, even when the fast policy, the LLM and the critic all agree that a command is fine and the operator is busy elsewhere?

A. The slow LLM
B. The critic
C. The formal safety shield
D. The digital twin

Core

Q6. LA-Light reports 20.4 % less waiting than RL. If (invented) RL waiting were 60 s, what would LA-Light's be?

A. 39.6 s
B. 47.8 s
C. 48.0 s
D. 72.2 s

Q7. In R1, talking cut delay from 18.45 to 17.79 s, but the message runs ranged over 0.67 s. What is the honest conclusion?

A. Talking clearly helps by 3.6 %
B. Latency clearly hurts
C. The effect is within single-seed noise; test more seeds
D. Messages make things worse

Q8. In R2 with 0 % bad answers, the unshielded agent had less delay (9.02 s) than the shielded one (10.62 s). Why?

A. The shield is badly coded
B. It broke minimum green 112 times
C. It had fewer vehicles
D. It used a better LLM

Q9. In R2 at 40 % bad answers, delay was 12.14 s with the shield and 31.49 s without. What is the reduction, relative to the unshielded run?

A. 19.4 %
B. 38.6 %
C. 61.4 %
D. 159 %

Q10. Which experiment probes the mixed-autonomy gap?

A. R1
B. R2
C. R3
D. R4

Q11. In R4, a fairness weight of 0.15 changed the longest wait from 53 s to 22 s. What was the relative change?

A. −31 %
B. −41.5 %
C. −58.5 %
D. −141 %

Q12. Feng et al. (2015) found connected-vehicle adaptive control cut delay up to 16.33 % at high penetration. At low penetration it was:

A. Even better
B. About the same as actuated control
C. Unsafe
D. Not tested

Q13. What does the 'sim-to-real' gap mean?

A. Simulators are too slow
B. Performance changes when a controller moves from the simulator to real streets
C. Real streets have more lanes
D. SUMO and CityFlow disagree

Q14. In the layered design, what is the digital twin used for?

A. Replacing detectors
B. Running new policies in shadow mode and measuring sim-to-real gaps
C. Talking to drivers
D. Enforcing minimum green

Challenge

Q15. 2,510 lights each decide every 5 s, and 1 % of decisions go to an LLM. About how many LLM calls per second?

A. 0.5
B. 5
C. 50
D. 502

Q16. R5: at H = 8 s delay rose from 16.72 to 21.12 s and CO2 from 229.5 to 241.4 g. Which is right?

A. Delay +4.4 %, CO2 +11.9 %
B. Delay +26.3 %, CO2 +5.2 %
C. Delay +20.8 %, CO2 +4.9 %
D. Delay −26.3 %, CO2 −5.2 %

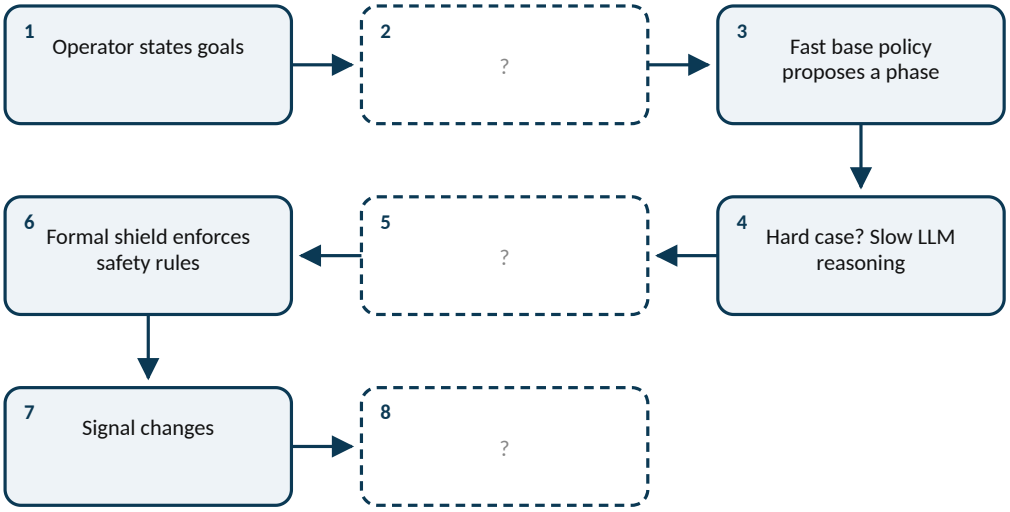
Q17. Which open question asks whether an LLM's stated reason matches what actually drove its decision?

A. Verified supervision
B. Faithful explanations
C. Scalable escalation
D. Certification

Q18. Three seeds give a new controller 10, 11, 12 s and the old one 11.5, 12, 12.5 s (invented). What should you report?

A. 'The new one is 1 s better', nothing more
B. Means 11 and 12 s with ranges 2 and 1 s: promising but not shown; run more seeds
C. The old one is better
D. Only the best seed

Draw it. Fill the blank boxes of this flow from memory (it is Figure 5).



Where to go if you missed a question

Missed	Do this
Q1, Q4, Q6	Re-read concept 1: The research landscape: six waves
Q3, Q7, Q8, Q18	Re-read concept 2: How to evaluate a new idea fairly
Q2, Q11, Q12, Q13	Re-read concept 3: The eleven gaps
Q9, Q10	Re-read concept 4: Our five experiments, R1-R5
Q5, Q14, Q15	Re-read concept 5: Future solution: a layered agentic controller
Q16, Q17	Re-read concept 6: Open research questions

Make it stick

BUILD THE SMALLEST WORKING EXAMPLE (15 min)

On paper, make a small results table for R2 (with and without shield, at 0 % and 40 % bad answers: delay and violations). Compute the relative change in delay at each level. Then write one sentence on why the 0 % row is misleading.

Self-check: 0 %: shield 10.62 s vs 9.02 s (+17.7 %, but 112 violations without). 40 %: 12.14 vs 31.49 s (61.4 % less with the shield; 0 vs 223 violations). The 0 % row looks bad for the shield only because the unshielded agent breaks minimum green.

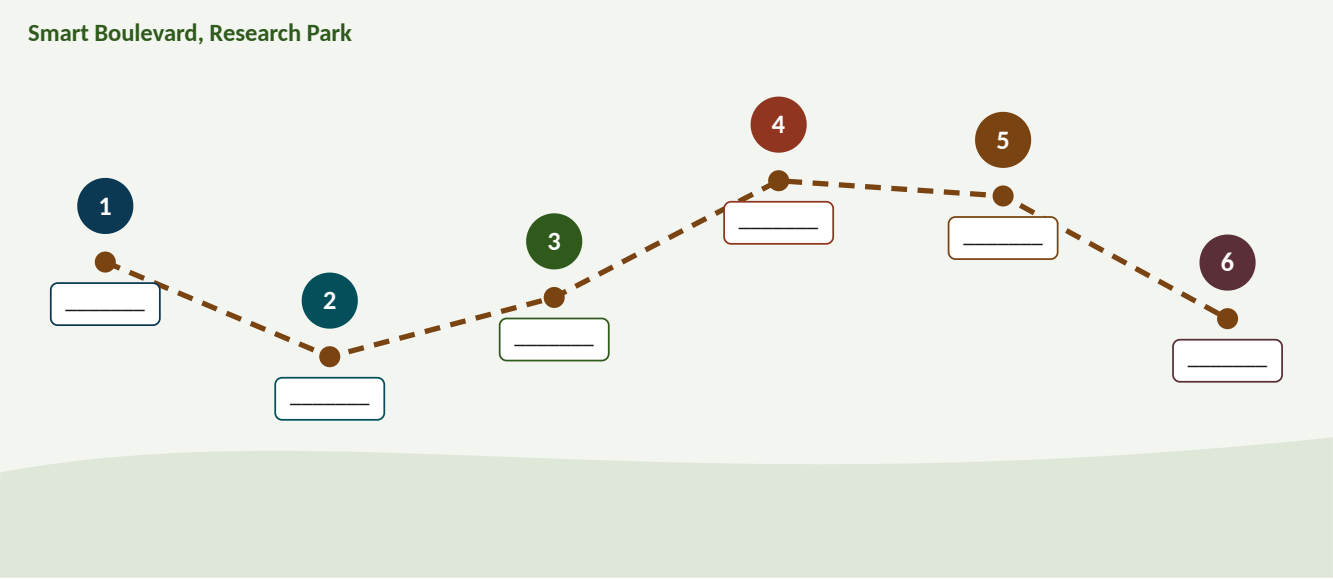
TEACH IT TO A 14-YEAR-OLD (10 min)

Explain to a 14-year-old why a clever AI that sometimes makes things up should not be allowed to switch traffic lights on its own, and what we put in between.

Words you may not use: hallucination, shield, LLM, reinforcement, sim-to-real

Fill the blank route

Write the concept held at each stop, and its key number.



One-page cheat sheet

1. The research landscape: six waves

Max-pressure 2013: local queues, proven max throughput. MPLight: 2,510 lights. LLM gains 17.5 % (iLLM-TSC), 20.4 % (LA-Light), 20.03 % (HeraldLight). 0 of 9 LLM papers report field deployment. new = old × (1 − g/100).

Route: Museum of waves

2. How to evaluate a new idea fairly

Compare with actuated and max-pressure. ≥ 3 seeds; mean and spread. R1: no messages 18.45 s, talking 17.79 s (−0.66 s), message runs range 0.67 s, mean 17.86 s: within noise. R2 0 % bad: no shield 9.02 s but 112 violations.

Route: Race track

3. The eleven gaps

Gaps with our experiments: robustness R1, safety + hallucination R2, mixed autonomy R3, equity R4, predictability R5. R4: w 0 → 0.15: longest wait 53 → 22 s (−58.5 %), delay 9.69 → 9.56 s. MP longest wait 150 s. Feng 2015: −16.33 % at high penetration.

Route: Pothole road

4. Our five experiments, R1-R5

R1: 18.45 vs 17.79 s. R2 at 40 %: 12.14 vs 31.49 s, 0 vs 223 violations, longest wait 39 vs 132 s. R2 at 0 %: 10.62 vs 9.02 s (112 violations). R3 10 %: 9.94 vs 9.01 s; 100 %: 6.6 s. R4: 53 → 22 s.

Route: Five lab rooms

5. Future solution: a layered agentic controller

Seven layers: base, LLM, critic, shield, twin, V2X, operator. VLMLight: < 1 % normal-condition degradation, up to 65 % less emergency waiting. 2,510 lights × 720 = 1,807,200 decisions/h; 1 % → 18,072 LLM calls/h.

Route: Layer-cake café

6. Open research questions

8 questions: verified supervision, faithful explanations, sim-to-real gap, combined faults, escalation, equity, < 25 % CAVs, certification. R5 H 0/4/8/12/16: delay 16.72/18.87/21.12/25.52/30.10 s; CO2 229.5/237.6/241.4/252.0/262.3 g. Fixed + GLOSA 27.22 s, 250.1 g.

Route: Lighthouse

Numbers to remember

- LLM gains reported: 17.5 % (iLLM-TSC), 20.4 % (LA-Light), 20.03 % (HeraldLight); 0 of 9 in the field
- MPLight: 2,510 lights; Feng 2015: −16.33 % at high CV penetration
- R1: 18.45 s silent vs 17.79 s talking; runs range 0.67 s (noise)
- R2 at 40 % bad: 12.14 vs 31.49 s; 0 vs 223 violations
- R3 at 10 %: CAV-only 9.94 s vs loops 9.01 s; 100 %: 6.6 s
- R4: w = 0.15 → longest wait 53 → 22 s; max-pressure 150 s
- R5: H = 8 s fewest stops 0.86; delay 16.72 → 21.12 s, CO_2 229.5 → 241.4 g

Answer key

Try it now

Try it 1 (The research landscape: six waves): 40 × (1 − 0.65) = 40 × 0.35 = **14 s**, a saving of 26 s. Remember 'up to': this is the best case the authors report.

Try it 2 (How to evaluate a new idea fairly): New: mean 33 ÷ 3 = **11 s**, range 2 s. Old: mean 36 ÷ 3 = **12 s**, range 1 s. The 1 s gain is **smaller than the new controller's 2 s spread**: promising, but run more seeds before claiming it.

Try it 3 (The eleven gaps): Half of 25 % = **at least 12.5 %** (in the authors' simulation). The gap: below 25 % penetration the benefit is much less certain.

Try it 4 (Our five experiments, R1-R5): (9.94 − 9.01) ÷ 9.01 × 100 = **+10.3 %** worse at 10 %; **0 %** at 100 %. Low penetration is where estimating unequipped vehicles matters most.

Try it 5 (Future solution: a layered agentic controller): 3600 ÷ 4 = 900 per signal; 400 × 900 = **360,000** decisions per hour; 360,000 × 0.02 = **7,200** LLM calls per hour (2 per second).

Try it 6 (Open research questions): Example: 'Can probabilistic SPaT with commitment windows beat actuated control on a corridor with 10-50 % CAVs, measured by CO_2 per vehicle, over 3 seeds?' Delay change: (18.87 − 16.72) ÷ 16.72 × 100 = **+12.9 %**.

Self-quiz

Q1: C. Adaptive systems date from the 1980s; max-pressure 2013; deep RL about 2019; LLMs from 2023. (Concept 1)

Q2: B. Hallucination is confident wrong output. Slowness is latency; lost messages and broken detectors are robustness problems. (Concept 3)

Q3: B. A seed sets the random arrivals. Several seeds show whether a gain is bigger than random variation. (Concept 2)

Q4: A. All were evaluated in simulation only, which is the sim-to-real gap. (Concept 1)

Q5: C. The shield holds the hard rules (minimum green, intergreen, conflicts). The critic advises and can be wrong. (Concept 5)

Q6: B. 60 × (1 − 0.204) = 60 × 0.796 = 47.76 ≈ 47.8 s. 72.2 s adds instead of subtracting. (Concept 1)

Q7: C. A 0.66 s gain is no bigger than the 0.67 s wobble, so one seed cannot show it. (Concept 2)

Q8: B. Breaking safety rules looks faster. Fair comparisons apply the same rules to every method. (Concept 2)

Q9: C. (31.49 − 12.14) ÷ 31.49 = 19.35 ÷ 31.49 = 61.4 %. 159 % divides by the shielded value instead. (Concept 4)

Q10: C. R3 controls with connected-vehicle data only, at 10-100 % penetration. (Concept 4)

Q11: C. (22 − 53) ÷ 53 = −31 ÷ 53 = −58.5 %. −31 is the change in seconds, not percent. (Concept 3)

Q12: B. Benefits depend on penetration; at low penetration it performed about like actuated control. This is the mixed-autonomy gap. (Concept 3)

Q13: B. Sim2Signal splits it into observation, action, transition and reward gaps; direct transfer always degraded. (Concept 3)

Q14: B. A calibrated simulation where policies are tested before going live. (Concept 5)

Q15: B. 2,510 × 720 = 1,807,200 decisions/h; 1 % = 18,072/h; ÷ 3600 ≈ 5 per second. 502 is all decisions per second. (Concept 5)

Q16: B. 4.40 ÷ 16.72 = 26.3 %; 11.9 ÷ 229.5 = 5.2 %. Option C divides by the new values. (Concept 6)

Q17: B. Faithfulness: a plausible reason may not be the real cause. Measuring it with engineers is open. (Concept 6)

Q18: B. The 1 s gain is smaller than the new controller's 2 s spread. Report mean and spread, and be cautious. (Concept 2)

Draw it: the completed flow is Figure 5.

Your revision plan

When	What	Done
Today	Read one concept a sitting. Do every Try-it-now on paper.	☐
Day 1	Walk the route from memory. Redraw the master mind map.	☐
Day 3	Warm-up and core quiz without notes. Re-read what you missed.	☐
Day 7	Whole quiz. Do the 'build the smallest example' task.	☐
Day 14	Teach-it task out loud. Cheat sheet from memory, then compare.	☐
Before an exam or project	Cheat sheet plus route walk only.	☐

Going further

- Varaiya, P. (2013). Max pressure control of a network of signalized intersections. *Transportation Research Part C: Emerging Technologies*. https://doi.org/10.1016/j.trc.2013.08.014
- Lai, S., Xu, Z., Zhang, W., Liu, H. and Xiong, H. (2025). LLMLight: Large Language Models as Traffic Signal Control Agents. *Proc. ACM SIGKDD (KDD 2025)*; arXiv:2312.16044. https://arxiv.org/abs/2312.16044
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HONESTY NOTE

R1-R5: small synthetic SUMO scenarios, one seed each, so small differences (notably R1) are within noise; the R2 'LLM' is a mock, so R2 tests the shield, not a real model. Literature results are as reported by their authors, in simulation only. 'Invented' examples use made-up baselines.